

TANJIM AHMED

EEE UNDERGRADUATE | RTL DESIGN | DESIGN VERIFICATION | ASIC / VLSI

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PROFESSIONAL SUMMARY

Electrical and Electronic Engineering undergraduate at AIUB (expected December 2026) with hands-on experience in RTL design, SystemVerilog verification, logic synthesis, and ASIC physical-design flow. Completed nine VLSI/RTL sub-projects and a modular smart ticket vending machine using Cadence Xcelium, Genus, and Innovus. Seeking an internship or entry-level opportunity in RTL Design, Design Verification, or Semiconductor/VLSI engineering.

TECHNICAL SKILLS

HDL & Programming: Verilog, SystemVerilog, C, C++, Python, C-shell scripting

EDA & Engineering Tools: Cadence Xcelium, Genus, Innovus; Xilinx Vivado; Silvaco TCAD

Design & Verification: RTL coding, FSM/datapath design, functional simulation, waveform debugging, class-based testbenches, constrained-random verification, assertions, functional coverage, synthesis, place-and-route, timing/power/area analysis

Platforms & Languages: Linux/Ubuntu

PROJECTS

Smart Ticket Vending Machine ([Smart Ticket Vending Machine](#)) | SystemVerilog | June 2026 – July 2026

- Designed modular RTL for station-based route validation and fare calculation, cash-denomination checking, payment tracking, invalid/fake-note rejection, change calculation, and cash-inventory management.
- Developed the SystemVerilog design and simulation testbench, debugged console and waveform issues, and verified transaction behavior using Cadence Xcelium.

RTL, Verification & ASIC Design Portfolio ([RTL & ALU Design](#) | [Datapath & Verification](#) | [RISC-V3 ASIC Design](#)) | GPD045 & Sky130 | May 2026 - July 2026

- Designed and verified 4-operation and registered 16-operation ALUs with arithmetic/logic, shift/rotate functions, status flags, synchronous reset, and SDC timing constraints.
- Performed synthesis, multi-Vt optimization, and PDK-based timing, leakage, switching-power, and area analysis; implemented pipelined polynomial datapaths with both parallel and resource-shared architectures.
- Built a class-based SystemVerilog verification environment for a 4x4 MAC using generator, driver, monitor, scoreboard, constrained-random stimulus, assertions, packages, and functional coverage.
- Synthesized and placed-and-routed a RISC_V3 processor; developed a hazard-free two-stage pipeline and a branch-enabled version with signed offsets and program-memory loading.

EDUCATION

Bachelor of Science in Electrical and Electronic Engineering (EEE) | American International University-Bangladesh (AIUB) 2023 - December 2026 (Expected) | Credits Completed: 128/148

Relevant Coursework: Digital Logic Design, VLSI Design, Microprocessor, Electronic Devices

Higher Secondary Certificate (HSC), Science

Rowmari Government College | 2020 |

Secondary School Certificate (SSC), Science

Rowmari C. G. Zaman Government High School | 2018 |

TRAINING & PROFESSIONAL DEVELOPMENT

- 9-Day Industrial Training Program - Ulkasemi Pvt. Limited | May 2026 | Design Verification, Analog Design, IC Physical Design, Foundry Technologies, and IC Mask Design.
- RTL to GDS-II Workshop - ProV Logic | 13-14 June 2026 | VLSI physical-design flow and complete RTL-to-GDSII methodology.
- VLSI Design and Semiconductor Industry Workshop - ChipXpert Technologies Pvt. Ltd. | 30-31 May 2026.